



Lean Software Development

SEng5304: Agile and Test Driven Development

Learning Objectives

Understand the Fundamentals of LSD

Understand the main LSD practices

Understand the Guiding Principles LSD

CHAPTER 05

AGENDA

Lean Software Development

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Introduction to Lean SW Dev

Defining Lean Software Development

Lean Software Development is an **Agile-aligned approach** focused on **maximizing value** for the customer by **eliminating waste and optimizing efficiency** throughout the software development process.

Rooted in Lean Manufacturing: It originates from the Lean manufacturing movement of the 1980s.

Maximum Customer Value: The main goal is to deliver the most value to the customer through efficient resource use and streamlined processes.

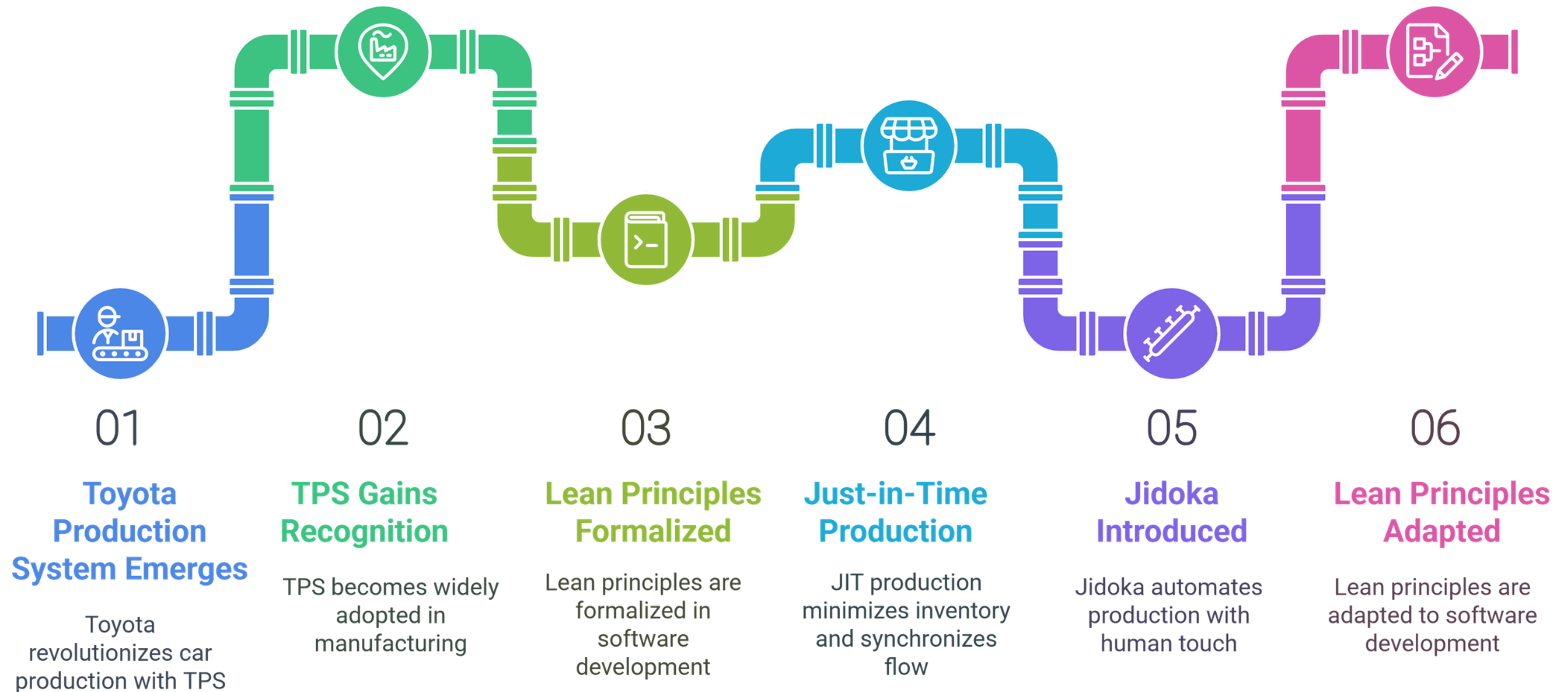
Eliminate Non-Value-Adding Elements: The core philosophy is to remove anything that doesn't directly contribute to the software product's value stream.

Prioritize High-Value Features: Development teams focus on delivering features most important to customers.

Lean aims to **improve speed and lower costs while also enabling responsiveness to evolving needs.**

Introduction to Lean SW Dev

Historical Journey



The Seven Guiding Principles of

LEAN SOFTWARE DEVELOPMENT

Eliminate Waste

Remove anything not adding product value. Unnecessary code, delays, and context switching.



Decide Late

Delay decisions until the last responsible moment.



Empower Team

Give teams autonomy, responsibility, and resources.



Optimize Whole

Optimize the entire value stream, concept to delivery.



Amplify Learning

Foster continuous learning through pair programming and knowledge sharing.



Deliver Fast

Focus on delivering value quickly and frequently.



Build Quality

Integrate quality assurance throughout development process.

Lean in Action

Practices and Methodologies

Eliminate Waste

- **Value Stream Mapping**: Identify and remove non-value-adding activities.
- **Just-In-Time Development**: Build features only when needed, reducing inventory (unfinished work).
- **Limit Work-in-Progress (WIP)**: Use Kanban to manage and visualize flow, minimizing multitasking and context switching.

Build Quality In

- **Test-Driven Development (TDD)**: Write tests before code to ensure correctness and reduce bugs.
- **Automated Testing**: Quickly detect defects to maintain a reliable codebase.
- **Continuous Integration (CI)**: Integrate and test code frequently to detect issues early.

Cont...

Create Knowledge

- **Pair Programming**: Encourage collaboration and real-time knowledge sharing.
- **Retrospectives**: Reflect on past iterations to gather insights and improve processes.
- **Documentation as Code**: Keep documentation lightweight and in sync with the software.

Defer Commitment

- **Set-Based Design**: Explore multiple design options and delay final decisions until the last responsible moment.
- **Prototyping and Experimentation**: Test hypotheses quickly before committing to large-scale changes.

Deliver Fast

- **Small Batch Sizes**: Deliver work in small, frequent increments for quicker feedback and reduced risk.
- **Short Iterations/Sprints**: Adopt time-boxed cycles (1–2 weeks) for rapid delivery and iteration.



Cont...

Respect People

- **Empowered Teams:** Allow teams to self-organize and make decisions.
- **Cross-Functional Teams:** Promote collaboration among developers, testers, designers, and business experts.

Optimize the Whole

- **Systems Thinking:** Focus on end-to-end process efficiency, not just local optimizations.
- **Feedback Loops:** Incorporate feedback at all stages (from customers, tests, and team members).

Key Benefits and Challenges

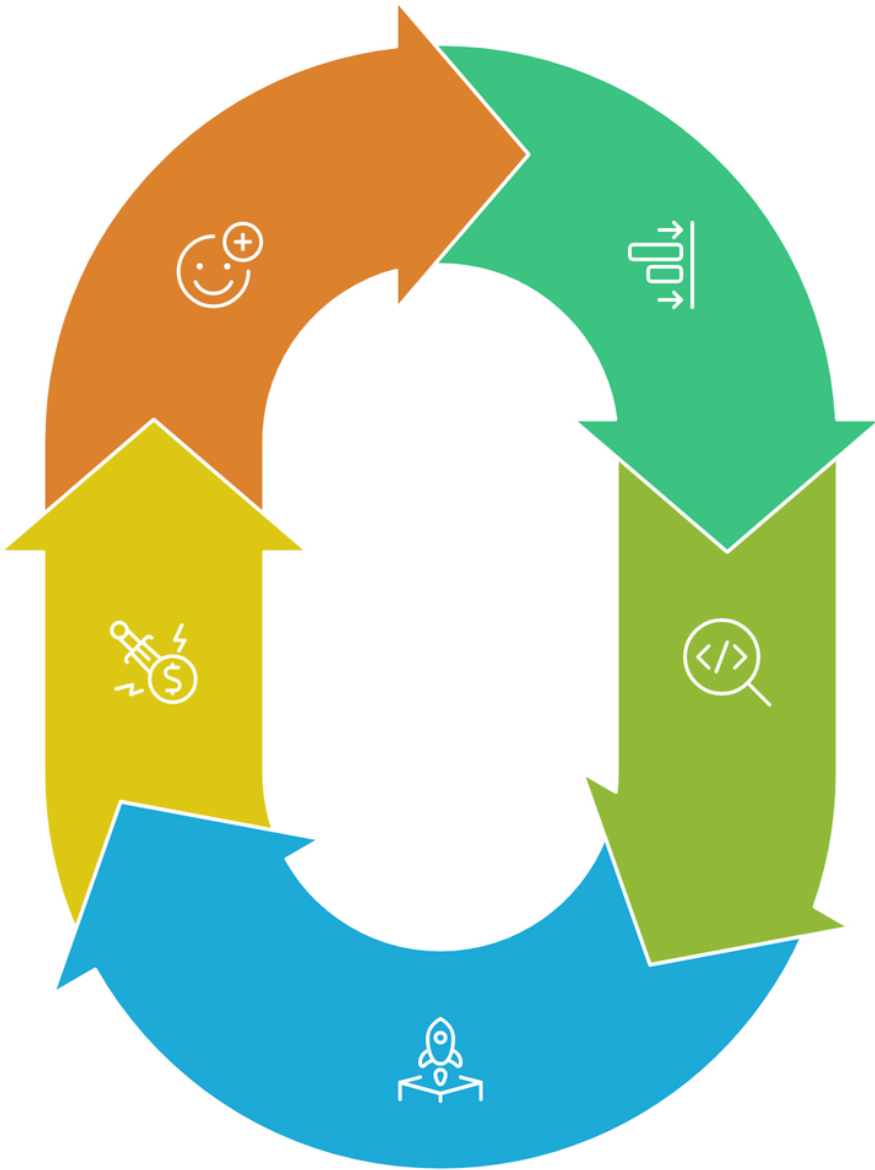
Key Benefits of Lean Adoption

Enhance Customer Satisfaction

By focusing on delivering valuable features frequently and incorporating feedback, teams are more likely to meet customer needs.

Reduce Costs

Eliminating waste, such as unnecessary features, delays, and rework, directly translates to lower development costs.



Improve Quality

A focus on building quality in throughout the development lifecycle reduces defects and rework, leading to more reliable software.

Increase Efficiency

By eliminating waste and streamlining processes, teams can achieve more with fewer resources and in less time.

Faster Delivery

Short development cycles and a focus on delivering value quickly get software into the hands of users sooner.

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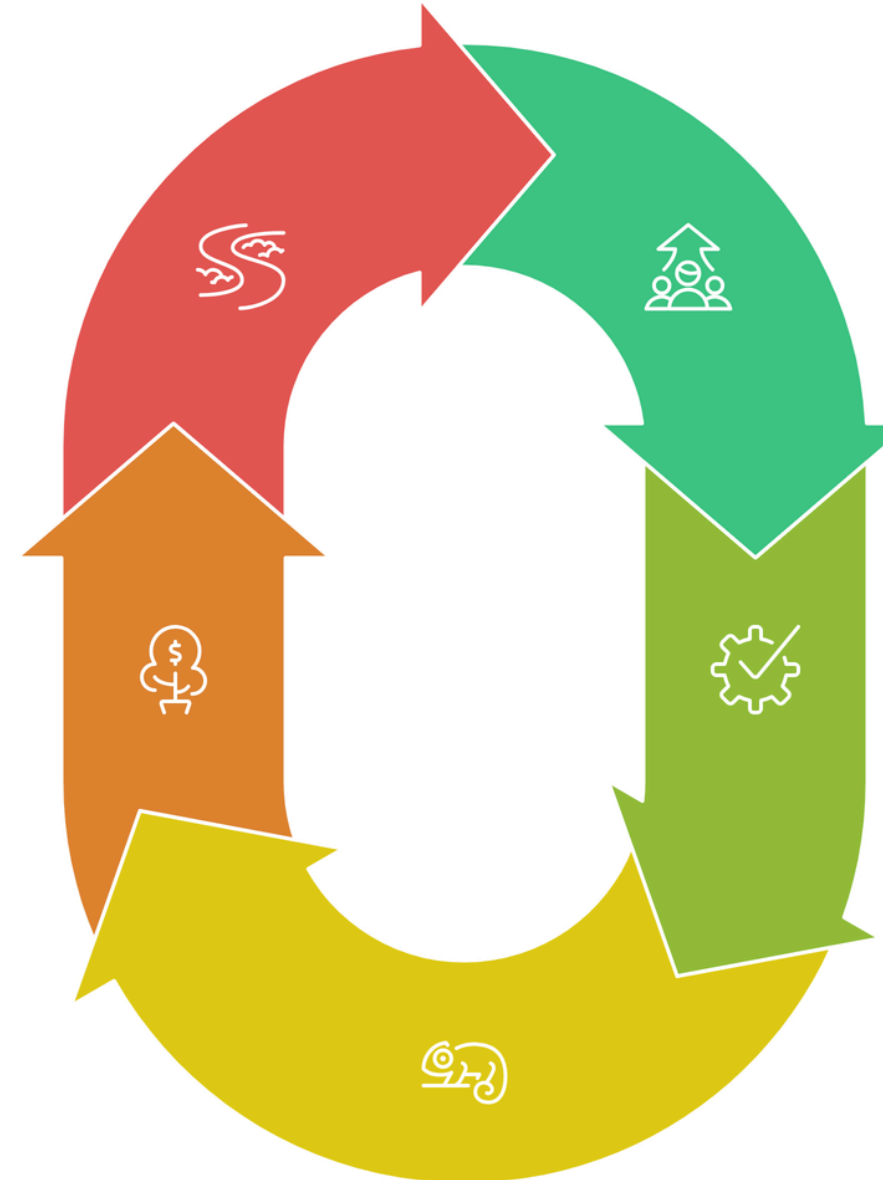
Key Benefits of Lean Adoption

Better Workflow

Streamlined processes and the elimination of bottlenecks lead to a smoother and more efficient workflow.

Sustainable Development

By focusing on long-term value and efficiency, Lean promotes sustainable development practices.



Empowered Teams

Lean encourages team autonomy and provides the necessary resources, leading to increased motivation and productivity.

Continuous Improvement

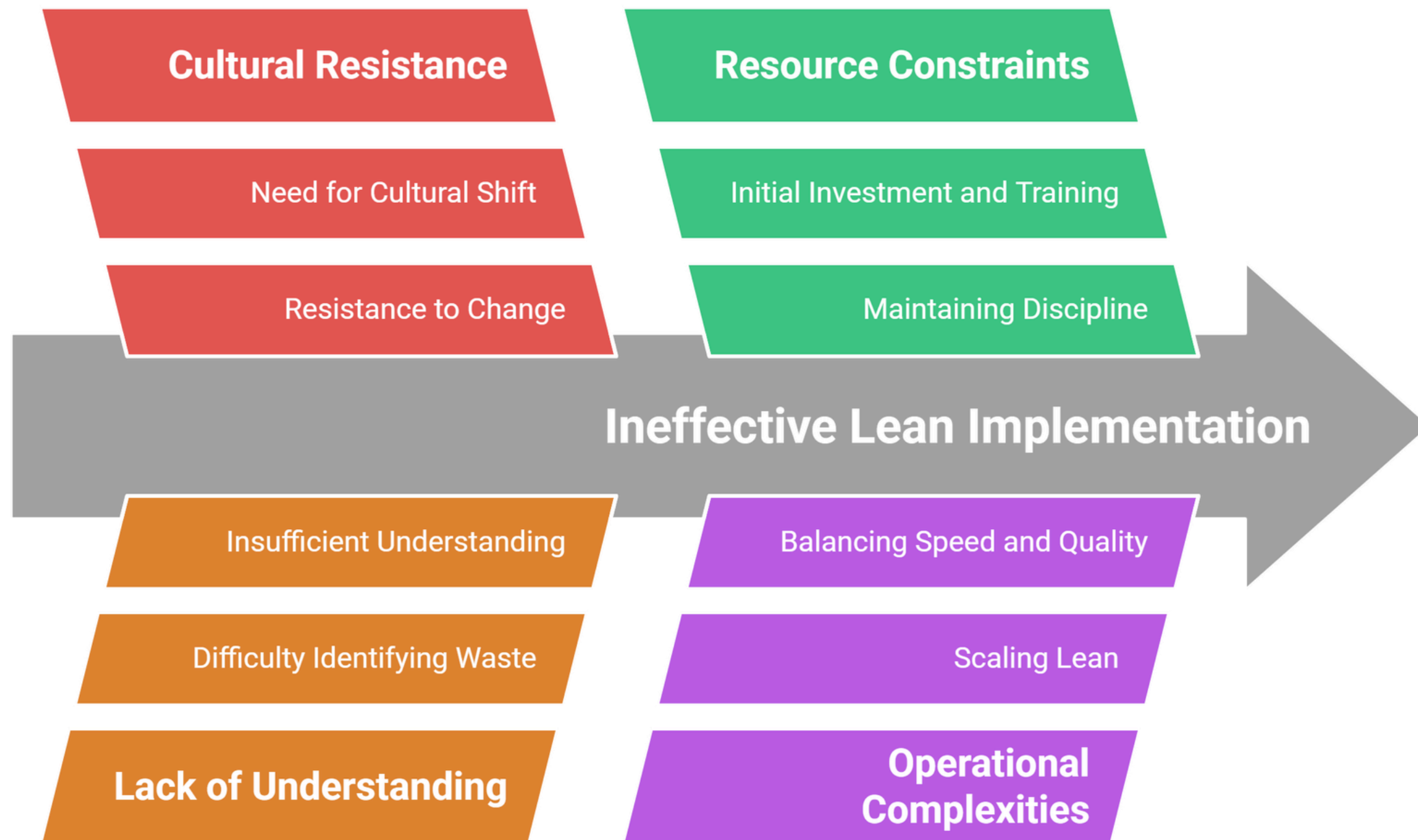
The principles of Lean foster a culture of learning and ongoing process optimization.

Flexibility and Adaptability

The ability to delay decisions and deliver frequently allows teams to respond effectively to changing requirements and market conditions.

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Key Challenges of Lean Adoption



The Lean Team: Roles and Responsibilities

While specific titles may vary, some common roles and their responsibilities include:

LEAN MASTER / LEAN COACH

Focus: Supporting a Lean mindset and systemic thinking.



Education and Support

Provides training and assistance in Lean principles

Waste Identification

Helps pinpoint and eliminate inefficiencies

Continuous Improvement

Encourages ongoing process enhancements

Facilitation

Leads retrospectives and learning sessions

Cont...

PRODUCT OWNER / CUSTOMER REPRESENTATIVE

Focus: Maximizing value and minimizing waste.



Customer Representation

Acts as the voice of the customer

Backlog Management

Defines and prioritizes the product backlog

Value Delivery

Ensures features deliver real customer value

Feedback Participation

Participates in continuous feedback loops

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DEVELOPMENT TEAM MEMBERS

Focus: "Build quality in" and deliver small, valuable increments fast.

Ownership

Team members feel responsible for their work

Cross-Functional Roles

Team members with diverse skills working together

Self-Management

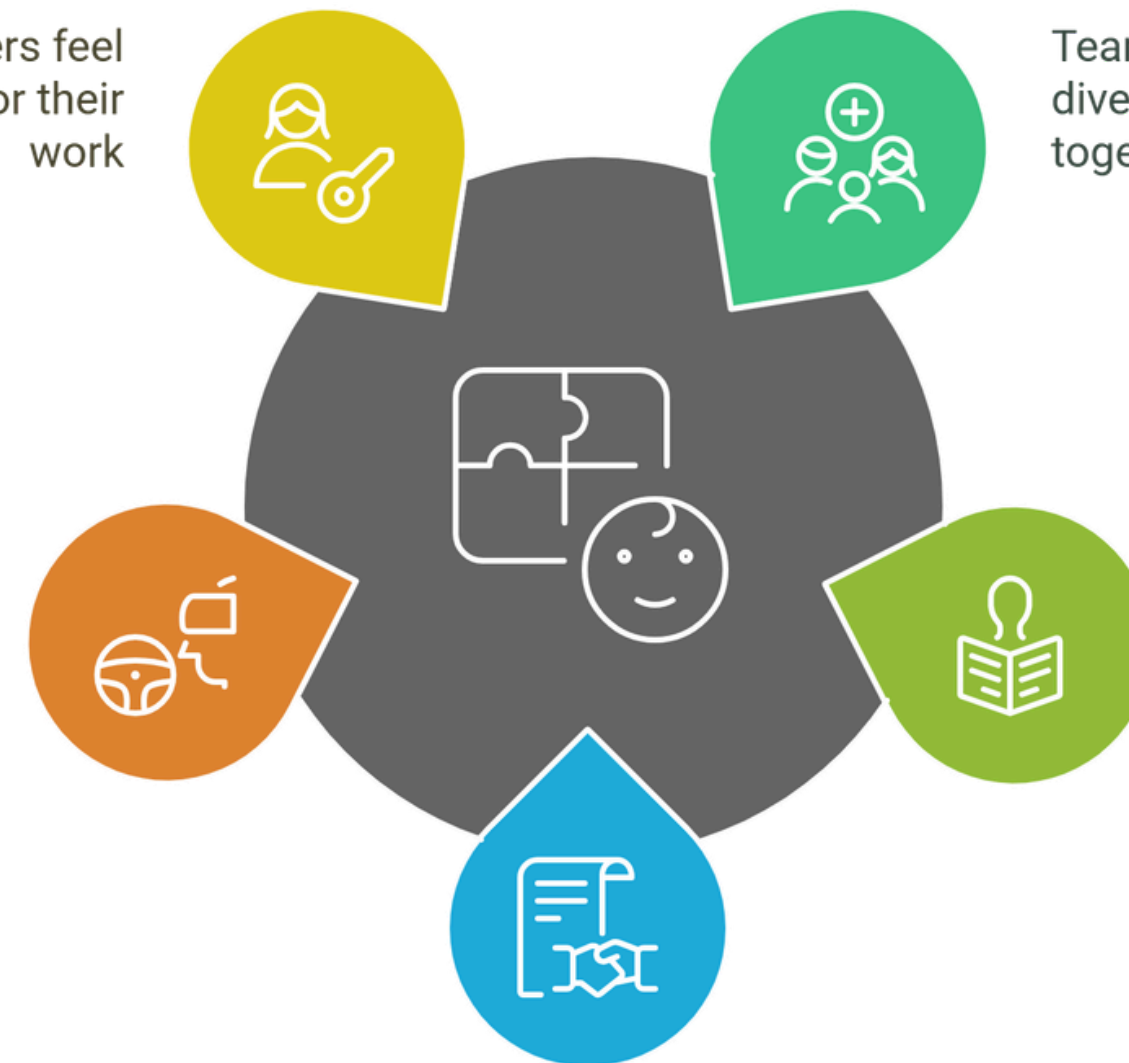
Team members are empowered to make decisions

Expertise

Team members are experts in their respective fields

Collaboration

Team members work together to design and implement solutions



End of Chapter 5

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